

ENERGY GLADIATORS - HACKATHON STRATEGY GUIDE

[CONFIDENTIAL] FOR YOUR EYES ONLY - DO NOT SHARE OR INCLUDE IN SUBMISSION

1. COMPETITION TIMELINE

When	What to Do
Friday 5pm	Kickoff - receive 2026 dataset + GitHub repo access
Friday Night	Clone official repo, start fresh project, upload recipe zip as REFERENCE only
Saturday (Full Day)	MAIN BUILD DAY - rebuild pipeline fresh, use AI Assistant live, document everything
Saturday Night	Polish notebook and slides, transfer to official templates, practice presentation
Sunday AM	Final push to GitHub by NOON, triple-check file names, practice timing (about 10 min)
Sunday PM	Presentations to judges, be ready for Q&A

2. STEP-BY-STEP EXECUTION PLAN

Phase 1: Setup (Friday Night, about 30 min)

- Clone official hackathon GitHub repo (NOT your personal repo)
- Upload recipe zip as a REFERENCE - DO NOT copy-paste code directly
- Start a NEW app.py - rebuild the structure from scratch
- This creates commit history showing genuine development

Phase 2: Data Exploration (Friday Night to Saturday AM)

- Upload NEW 2026 training/test data
- Use AI Assistant: "What patterns do you see in this data?"
- Document findings in notebook with markdown
- Create visualizations (histograms, box plots, correlation heatmap)

Phase 3: Build and Adapt (Saturday)

- Rebuild pipeline using recipe as guide (9 steps)
- Ask AI Assistant: "What features should I engineer?"
- Ask AI Assistant: "Would a different model work better?"
- DOCUMENT these AI conversations - show judges you are exploring

Phase 4: Polish (Saturday Night)

- Transfer work to official Jupyter notebook template
- Rename to EnergyGladiators.ipynb
- Add markdown explanations at EVERY step
- Create slides using official PowerPoint template
- ALL team member names on slide 1 (REQUIRED)

Phase 5: Submission (Sunday AM)

- Generate solution.csv with 100 realizations
- VERIFY: Columns are Real_1 through Real_100 (not R_1)
- Push everything to official GitHub repo
- Final commit BEFORE noon deadline
- Push 30 minutes early for safety buffer

3. THE 9-STEP WORKFLOW

Step	What It Does	Time Estimate
1. Data Upload	Load training (1,082 wells) and test (50 wells) CSVs	2 min
2. Data Cleaning	Handle missing values (median for numbers, mode for categories)	3 min
3. EDA	Visualize distributions, correlations, patterns	10 min
4. Feature Engineering	Create Time_Overrun, Total_Pumping_Time, Clusters_per_Stage	5 min
5. Model Training	Train Random Forest for Grid, Diesel, CNG with 5-fold CV	5 min
6. Uncertainty	Residual bootstrapping for 100 realizations	3 min
7. Generate Predictions	Create solution.csv	2 min
8. Quick Start	Reference instructions	-
9. AI Assistant	Ask questions, get help	Ongoing

Total workflow time: About 30 minutes once familiar

4. CRITICAL DOMAIN KNOWLEDGE

Fleet Type to Fuel Mapping

Fleet Type Uses Grid Uses Diesel Uses CNG

Grid	YES	NO	NO
Diesel	NO	YES	NO
Turbine	NO	NO	YES
DGB	NO	YES	YES

Key Insight: DGB (Dual-fuel) wells generate TWO output rows (one Diesel, one CNG). 50 wells become 63 rows in solution.csv (13 DGB wells x 2 = 26, plus 37 single-fuel wells)

5. YOUR KILLER DIFFERENTIATOR

Why Our Solution Beats Commercial Tools

“Commercial tools like Spotfire cost thousands per year and only show historical data. Our solution predicts future energy usage with uncertainty quantification - something even enterprise tools do not do. And our AI assistant means any engineer can adapt it to new problems without coding.”

Aspect	Commercial BI (Spotfire, Tableau)	Our Solution
Approach	Descriptive (what happened)	PREDICTIVE (what will happen)
Output	Charts and dashboards	Point estimates + uncertainty ranges
Cost	\$3,000-5,000/year license	Free (Python/Streamlit)
Adaptability	Fixed features	AI assistant adapts to any dataset
Skill Value	Vendor lock-in	Portable Python/ML skills

The Key Differentiator: - Spotfire = Reporting tool (“here is what your energy consumption was”) - Our solution = Intelligence tool (“here is what it will be, and how confident we are”)

Innovation Story for Judges

“We did not just build a model for the 2026 hackathon problem. We built a platform that can solve FUTURE problems. Ask it anything...”

Then do a LIVE DEMO - type a question like “What features would you suggest for this dataset?” and watch it respond.

6. AI ASSISTANT CAPABILITIES

The AI Assistant (Step 9) can now help with:

Team Onboarding

- “Walk me through how this app works step by step”
- “What does Step 5 do?”
- “Why do DGB wells have two rows?”

ML Guidance

- “What features should I engineer?”
- “Would XGBoost work better than Random Forest?”
- “Why is my model performing poorly?”

Local Installation

- “How do I run this on my laptop?”
- Full step-by-step instructions for Windows/Mac/Linux

Innovation Explanation

- “Why is this better than Spotfire?”
- “What makes this solution innovative?”

7. JUDGING CRITERIA - YOUR STRATEGY

Criterion	Weight	Your Strategy
Technical Accuracy	HIGH	Segmented models per fuel type (proven to work)
Reproducibility	20 pts	random_state=42 everywhere, clean notebook, requirements.txt
Code Quality	10 pts	Well-commented code, follow official template structure
Innovation	15 pts	AI ASSISTANT - live, adaptive, industry-agnostic ML helper
Interpretability	15 pts	Feature importance plots, domain-driven explanations
Presentation	HIGH	Live demo of AI Assistant answering questions

8. CRITICAL COMPLIANCE CHECKLIST

Requirement	Status	Action
Develop during hackathon	DONE	Rebuild fresh using recipe as guide
Use only 2026 official data solution.csv format	DONE	Train on new data only
	VERIFY!	Columns = Real_1 to Real_100 (not R_1)
Notebook = TeamName.ipynb	DONE	Rename to EnergyGladiators.ipynb
Slides = TeamName.pptx	DONE	All members on slide 1
Push to official GitHub	DONE	Use THEIR repo, not yours
100 realizations	DONE	Already implemented
Deadline: Noon Sunday	DONE	Push 30 min early for safety

9. FINAL REMINDERS

DO NOT:

- Copy-paste code from recipe (looks like pre-work)
- Use 2025 data for training (only 2026 official data)
- Forget team member names on slide 1
- Push to personal repo instead of official hackathon repo
- Include this PDF in any submission

DO:

- Rebuild pipeline fresh - use recipe as reference only
- Use AI Assistant LIVE during hackathon (innovation demo)
- Document your thought process in notebook markdown
- Create commit history showing genuine development
- Practice your presentation (aim for about 10 minutes)
- Demo the AI Assistant answering a live question

10. RECIPE ZIP CONTENTS (Reference)

File	Purpose
app.py	Complete Streamlit app with 9-step workflow + AI Assistant
notebooks/EnergyGladiators.ipynb	Jupyter notebook template
presentations/EnergyGladiators.md	Slide outline with innovation story
requirements.txt	Python dependencies
data/*.csv	2025 sample data (structure reference only)
docs/HACKATHON_CHECKLIST.md	Submission checklist

11. QUICK REFERENCE - PRESENTATION TALKING POINTS

Opening: “We are Team Energy Gladiators. We built an intelligent ML system that predicts energy consumption for hydraulic fracturing operations - with uncertainty quantification.”

Innovation Moment: “But here is what makes us different. Most teams built a model. We built a platform. Watch this...” [Type into AI Assistant: “What features would you recommend for optimizing this model?”]

Closing: “Commercial tools like Spotfire cost thousands and only tell you what happened. Our solution tells you what WILL happen, how confident we are, and can adapt to any future problem. Thank you.”

Team Energy Gladiators - Good luck!